EP-1

Suppose that D is a domain and that $0 \notin D$. Show that there is a branch of $\log z$ in D if and only if $1/z$ has an antiderivative on D.

Hint: if $f'(z) = 1/z$ for all $z \in D$, consider the function $h(z) = z/\exp(f(z))$ and note that $\exp(z) = a$ has infinitely many solutions if $a \neq 0$. (Not to be turned in: observe that this result implies that $f(z) = 1/z$ does not have an antiderivative on $\mathbb{C} \setminus \{0\}$.)

Pf. ($\Longrightarrow$) If L is a branch of $\log z$ on D. By Definition 1.24, $L : D \to \mathbb{C}$ is continuous, and $\forall z \in D$: $L(z) \in \log(z)$. I.e., $\exp(L(z)) = z$ for all $z \in D$.

We can show that L is holomorphic with $L'(z) = 1/z$. Fix $z_0 \in D$. Since the differential of exp never vanishes ($\exp' = \exp$), and $\exp(L(z_0)) = z_0 \neq 0$, the inverse function theorem gives that L is holomorphic near z_0. Differentiating $\exp(L(z)) = z$ by the chain rule gives

$$\exp(L(z))L'(z) = 1 \quad \text{i.e.} \quad L'(z) = \frac{1}{\exp(L(z))} = \frac{1}{z}.$$

Therefore L is an antiderivative of $1/z$ on D.

($\Longleftarrow$) If $f \in H(D)$ satisfies $f'(z) = 1/z$ for all $z \in D$. Using the hint, we let

$$h(z) = \frac{z}{\exp(f(z))}.$$

Since $\exp(f(z)) \neq 0$ for all z, the function h is holomorphic on D. Differentiate with the quotient rule:

$$h'(z) = \frac{\exp(f(z)) - z\exp(f(z))f'(z)}{\exp(f(z))^2} = \frac{\exp(f(z)) - z\exp(f(z))\frac{1}{z}}{\exp(f(z))^2} = \frac{\exp(f(z)) - \exp(f(z))}{\exp(f(z))^2} = 0.$$

Since D is a domain (open and connected set), h is constant: $h(z) = c$ for some $c \in \mathbb{C}$. Thus $\forall z \in D : z = c\exp(f(z))$. Since $0 \notin D$, we have $c \neq 0$.

By Proposition 1.22, since $c \neq 0$, the equation $\exp(w) = c$ has solutions (infinitely many, differing by multiples of $2\pi i$). Fix any $a \in \mathbb{C}$ with $\exp(a) = c$. Then $\forall z \in D : z = \exp(a)\exp(f(z)) = \exp(f(z) + a)$. This means $f(z) + a \in \log(z)$ for every $z \in D$.

Define $L(z) = f(z) + a$. Then L is holomorphic (thus continuous) and $L(z) \in \log(z)$ for all $z \in D$, so L is a branch of $\log z$ on D by Definition 1.24.

Thus shown that there is a branch of $\log z$ in D iff $1/z$ has an antiderivative in D. □

Ex 3.2.2

Exercise 3.2.2. Let $\gamma : [0,1] \to \mathbb{C}$ be a Jordan arc. Show that the set γ^* has no interior. (In particular, Jordan arcs are far from "space-filling".)

Pf. From Definition 3.3, a Jordan arc is an injective curve and $\gamma^* = \gamma([0,1])$ denotes its image.

Since γ is a continuous bijection from the compact set $[0,1]$ onto γ^*, γ is a homeomorphism from $[0,1]$ to γ^*.

Suppose, for contradiction, that γ^* has nonempty interior. Then there exists an open disk $D_r(w) \subseteq \gamma^*$. Since $D_r(w)$ contains more than two points, we can choose $w' \in D_r(w)$ with $p := \gamma^{-1}(w') \in (0,1)$. Pick $r' > 0$ so that $D_{r'}(w') \subseteq D_r(w) \subseteq \gamma^*$.

Since γ is a homeomorphism, removing w' from γ^* gives

$$\gamma^* \setminus \{w'\} \cong [0,1] \setminus \{p\} = [0,p) \cup (p,1],$$

which is disconnected. So $\gamma^* \setminus \{w'\}$ has exactly two connected components: $\gamma([0,p))$ and $\gamma((p,1])$.

On the other hand, the modified disk $D_{r'}(w') \setminus \{w'\}$ is connected (it is path-connected: any two points can be joined by an arc in the modified disk). Since it is a connected subset of $\gamma^* \setminus \{w'\}$, it must lie entirely in one component, say $\gamma([0,p))$.

But γ is continuous and $\gamma(p) = w'$, so for sufficiently small $\varepsilon > 0$ we have $|\gamma(p+\varepsilon) - w'| < r'$, meaning $\gamma(p+\varepsilon) \in D_{r'}(w')$. Since γ is injective and $p+\varepsilon \neq p$, we have $\gamma(p+\varepsilon) \neq w'$, so $\gamma(p+\varepsilon) \in D_{r'}(w') \setminus \{w'\} \subseteq \gamma([0,p))$. But $p+\varepsilon \in (p,1]$ means $\gamma(p+\varepsilon) \in \gamma((p,1])$, which is disjoint from $\gamma([0,p))$ by injectivity, which is a contradiction. Therefore γ^* has no interior. □

Ex 3.2.5

Let S be a nonempty subset of the plane. Let $\mathrm{Conv}(S)$ be the collection of sums of the form $\sum_{k=1}^{n} t_k z_k$ such that $n \in \mathbb{N}$, each $t_k \geqslant 0$, each $z_k \in S$, and $\sum_{k=1}^{n} t_k = 1$. Show that $\mathrm{Conv}(S)$ is the convex hull of S; that is, show that $\mathrm{Conv}(S)$ is the smallest convex subset of the plane containing S.

Pf. To show Conv(S) is the convex hull of S, we show that:

1. $S \subseteq \mathrm{Conv}(S)$: $\forall z \in S$, take $n = 1$ and $t_1 = 1$. Then $z = (1)(z) \in \mathrm{Conv}(S)$.

2. $\mathrm{Conv}(S)$ is convex:

   Take $a, b \in \mathrm{Conv}(S)$ and $\lambda \in [0,1]$. Let $a = \sum_{k=1}^{n} t_k z_k$ and $b = \sum_{j=1}^{m} s_j w_j$ where each $t_k, s_j \geqslant 0$, each $z_k, w_j \in S$, and $\sum t_k = \sum s_j = 1$ (as in the question). Then

   $$\lambda a + (1-\lambda) b = \sum_{k=1}^{n} (\lambda t_k) z_k + \sum_{j=1}^{m} ((1-\lambda) s_j) w_j.$$

   Every coefficient $\lambda t_k \geqslant 0$ and $(1-\lambda) s_j \geqslant 0$, and each $z_k, w_j \in S$. Thus the coefficients sum to

   $$\lambda \sum_{k=1}^{n} t_k + (1-\lambda) \sum_{j=1}^{m} s_j = \lambda + (1-\lambda) = 1.$$

   Hence $\lambda a + (1-\lambda) b \in \mathrm{Conv}(S)$, and $\mathrm{Conv}(S)$ is convex.

3. $\mathrm{Conv}(S)$ is contained in every convex set that contains S:

   Let C be any convex set with $S \subseteq C$. We show by induction on n that every element $\sum_{k=1}^{n} t_k z_k \in \mathrm{Conv}(S)$ belongs to C.

   For the base case, take $n = 1$: we have $t_1 = 1$ and $z_1 \in S \subseteq C$, so the sum $z_1 \in C$.

   Then consider the result for convex combinations of $n-1$ points of S, and $\sum_{k=1}^{n} t_k z_k$ with each $t_k \geqslant 0$, each $z_k \in S$, and $\sum t_k = 1$. If $t_n = 1$, then all other $t_k = 0$ and the sum equals $z_n \in C$. Otherwise $t_n < 1$, then

   $$\sum_{k=1}^{n} t_k z_k = (1 - t_n) \underbrace{\sum_{k=1}^{n-1} \frac{t_k}{1-t_n} z_k}_{=: w} + t_n z_n.$$

   Since $\sum_{k=1}^{n-1} \frac{t_k}{1-t_n} = \frac{1-t_n}{1-t_n} = 1$ and each $\frac{t_k}{1-t_n} \geqslant 0$, the point w is a convex combination of $n-1$ points of S. By the inductive hypothesis, $w \in C$. Since $z_n \in S \subseteq C$ and C is convex, we get $(1-t_n) w + t_n z_n \in C$. By induction, $\mathrm{Conv}(S) \subseteq C$.

   Since C was an arbitrary convex set containing S, $\mathrm{Conv}(S)$ is the smallest convex set containing S, i.e., the convex hull of S. □